Table 4.7 Nutritive value of milk and milk products (per 100 g)					
Food	**Energy (Kcal)**	**Protein (g)**	**Fat (g)**	**Calcium (mg)**	**Iron (mg)**
Cow's milk	67	3.2	4.1	120	0.2
Buffalo's milk	117	43	6.5	210	0.2
Curd	60	3.1	4.0	149	0.2
Paneer	265	18.3	20.8	208	-

is 8.5 percent. It is prepared from a mixture of buffalo milk and skimmed milk.

4. Homogenised Milk

Homogenisation is a mechanical process that reduces the size of fat globules by forcing milk through small apertures under pressure and velocity. When milk is homogenised, the average size of the globule will be 2 micrometers. The decrease in the size of fat globules increases their numbers and surface area. The newly formed fat droplets brings about stabilization of the milk emulsion and thus prevents rising of the cream. Homogenised milk has a creamier texture, bland flavour and whiter appearance.

5.Evaporated Milk

It is made by evaporating more than half the water from milk under vacuum, at a temperature of 74°C- 77°C. It is then fortified with vitamin D, homogenized and filled into cans.

4.5.3 Pasteurisation of milk

Milk is a favourable medium for bacterial growth. Pasteurisation destroys all pathogenic bacteria, including those causing typhoid, tuberculosis, diphtheria as well as yeasts and moulds. Pasteurization is a process which consists of heating milk to a certain temperature for a definite time to ensure destruction of harmful bacteria. There are three methods of pasteurisation.

a) **Holding method or Batch process :** In this method, milk is held at 62.8°C for 30 minutes.

b) **High temperature short time method or continuous process (HTST):** Milk is heated to 71.7°C for not less than 15 seconds.

c) **Ultra High temperature method :** Milk is heated to a temperature of 93.4°C for 3 seconds.

Milk should be then rapidly cooled to prevent multiplication of surviving bacteria, after heat treatment. During pasteurization the nutritive value of milk is not altered. It does not produce an unpleasant cooked flavour. Harmful pathogens especially TB bacteria are destroyed. Shelf life of milk is increased due to a marked decrease in the total bacterial count.

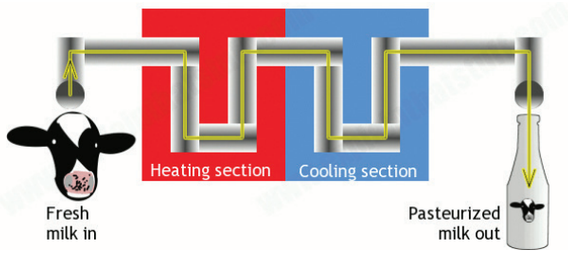


Fig 4.9: Pasteurisation of milk

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